AI251P Applied Computer Vision Capstone Project – Phase 1 Report

Smart Pet Vision: Cat and Dog Classification, Detection and Segmentation

Phase 1 (M1–M3, 40 points) · Due 28 October 2026

1. Team and Project

Field	Value
Team name	Smart Pet Vision
Topic	Original topic proposed. **Instructor approval:** Topic approval remains pending. Prior approval and eligibility for the original-topic bonus have not been established.
Application area	Animal welfare: preparing cat and dog photographs for adoption listings
Code repository	https://github.com/Yasser-20-alhajlah/smart-pet-vision Commit Hash: `faf05893291e9c1209cd07d63d891bc0b8acc2de`
W&B project	wandb.ai/yasser-alhajlah-kaust/smart-pet-vision; summary report: W&B report

Team Members

Name	Student ID	Phase 1 Contribution
Abdulrahman Mastour Alharthi	225478	Data cleaning, split and audits (M1); the three classifiers and their evaluation (M2–M3); final edit of this report and evidence folders.
Yasser Mansour Alhajlah	225481	Body boxes derived from trimaps; W&B tracking setup; independent reproduction on a clean runtime.
Mohammed Adeeb Alrashdi	225488	Masks derived from trimaps; annotation guide for the team photos.
Musaad Obaied Saad Alqahtani	225471	Repository structure, unified configuration; reproducibility instructions.

Note: All four members reviewed and corrected about 100 team photos each in CVAT.

Problem and Acceptance Criteria

- **Summary:** The system takes an adoption-listing photograph and determines whether it is ready for immediate publishing or requires referral to shelter staff. It reports species with a temperature-scaled confidence score, bounding body box, segmentation mask, cut-out animal image, and a publication verdict. Phase 1 establishes the empirical data foundations and the primary species classifier.
- **Input → Output:** Single cat or dog photograph → species, calibrated confidence, body box, segmentation mask, cut-out area share, and operational triage verdict.
- **Intended users and decision supported:** Animal shelter staff and adoption platform moderators. A photograph is referred when the predicted confidence falls below threshold, when the classifier and detector disagree on species, or when no animal is identified.
- **Acceptance criteria (fixed a priori):** Phase 1: Test accuracy ≥ 0.90 and test macro-F1 ≥ 0.90 for the selected classifier. Phase 2 (reference): mAP50 ≥ 0.75, recall ≥ 0.80, mIoU ≥ 0.75, Dice ≥ 0.85, latency ≤ 2 s on a T4 GPU.

2. Data Foundations (M1)

2.1 Datasets

Dataset	Link / Reference	Content and annotations	Use	License	Images
Oxford-IIIT Pet	Parkhi et al. (2012)	37 breeds (12 cat, 25 dog); species and breed labels, pixel trimaps, head boxes	Train/Val/Test	CC BY-SA 4.0	7,349
Team-collected	Private directory `SmartPetVision_TeamData`	Photos supplied by team and owners; species, derived boxes, CVAT polygon masks	External test benchmark only	Private consent	401

The full Oxford-IIIT Pet dataset is utilised; 14 unparseable empty-trimap instances were dropped, retaining 7,335 verified images.

2.2 Cleaning

Check	Method / Tool	Images removed
Corrupt or unreadable files	Pillow open & verify; minimum side 32 px	0
Exact duplicates	MD5 byte hash verification	0
Near-duplicates	pHash, Hamming distance ≤ 4 (41 clusters / 84 images kept intact within single splits)	0
Empty trimap	Absence of parseable mask or bounding region	14
Images remaining	**2,366 cats, 4,969 dogs across 37 breeds** (Median size 500 × 375 px)	**7,335**

2.3 Leakage-Safe Split

- **Grouping key:** Near-duplicate clusters via perceptual hashing (Hamming distance ≤ 4, transitive union-find), yielding 7,292 distinct clusters. Groups cross no splits (cross-split count = 0).
- **Stratification:** 70/15/15 ratio stratified by breed at the group level (seed 42).

Split	Groups	Images	Class Breakdown
Train	5,091	5,123	Cat: 1,652 (32.2%), Dog: 3,471
Validation	1,103	1,106	Cat: 357 (32.3%), Dog: 749
Test	1,098	1,106	Cat: 357 (32.3%), Dog: 749

> **Identity-leakage limitation:** Duplicate groups do not cross the train, validation and test splits. However, animal identities are unavailable, so identity-disjointness is not established. Image-similarity filtering is a sensitivity analysis, not a measurement or causal upper bound on leakage.

2.4 Label and Annotation Audit

- **Species audit (200 images):** 0 errors observed (0.0% point error; 95% one-sided upper bound: 1.5% via rule of three).
- **Breed audit (200 images):** 4 ambiguous labels identified (2.0%; 95% Wilson CI: 0.8%–5.0%).
- **Derived bounding box audit (300 images):** 4 loose boxes noted due to background inclusion (1.3%; 95% Wilson CI: 0.5%–3.4%). Zero animal omissions detected.

2.5 Data Loader and Augmentation

Input resolution: 224 × 224 px; ImageNet channel normalization; batch size 64. Augmentation pipeline: `RandomResizedCrop` (scale 0.5–1.0), horizontal reflection (p=0.5), and subtle `ColorJitter` (brightness/contrast 0.2, hue 0.02). Validation and test pipelines employ deterministic resize to 255 px followed by central cropping to 224 px.

3. Classification (M2–M3)

3.1 Model Architectures & Selection

Trained via AdamW (weight decay 10^{-4}), cosine annealing, mixed precision, cross-entropy loss, and seed 42. Selection protocol: Models evaluated strictly on validation macro-F1; the test set was locked until final evaluation.

Model	Backbone	Epochs	Val. Acc.	Test Accuracy	Test Macro-F1	ECE (Raw → Calibrated)
Scratch	Custom 10-layer CNN	25	0.928	90.5967%	0.892300	0.014 → 0.025
Frozen	Pretrained ResNet-50	8 (best 6)	0.998	99.9096%	0.998965	0.011 → 0.001
Fine-tuned	Pretrained ResNet-50	10 (best 6)	**1.000**	**99.9096%**	**0.998965**	0.0015 → 0.0009

> Frozen and fine-tuned models achieved equal aggregate test accuracy and macro-F1. McNemar testing detected no difference on this test set (exact p = 1.0); this does not establish equivalence. The fine-tuned model was selected using validation macro-F1, not test performance.

3.2 Learning-Curve Provenance

> **Learning-curve provenance:** Training accuracy and validation loss were measured in full-logging reruns. Shared training-loss, validation-accuracy and macro-F1 histories agree within 1e-12; runtimes differ. The additional curves are rerun measurements, not recovered original training logs. Submitted checkpoints were retained.

Observations: The custom scratch model shows slight under-training rather than overfitting at epoch 25 (training accuracy 0.914 vs validation accuracy 0.928). The fine-tuned architecture reaches minimal validation loss (0.0018) at epoch 6 (Val Acc: 100%), justifying early checkpoint preservation.

3.3 Per-Class Performance (Selected Fine-tuned Model)

Class	Precision	Recall	Macro-F1	Support
Cat	1.000	0.997	0.999	357
Dog	0.999	1.000	0.999	749

3.4 Calibration & Confidence Dynamics

The single misclassification by the fine-tuned model was an atypical Bombay cat predicted as a dog. Because the validation split was near-perfectly separable, temperature scaling converged to the predefined lower clipping boundary (T = 0.05). Consequently, confidence calibration on this split is mathematically ill-posed, and operational referral thresholds remain provisional until evaluated on a dedicated, non-separable calibration split.

3.5 Error Taxonomy & Failure Patterns

- **Hairless, black, and atypical coats:** Low-contrast black coats (Bombay) and atypical textures (Sphynx) represent the primary failure mode.
- **Extreme curling poses:** Postures obscuring typical anatomical landmarks induced classification challenges in the scratch model.

3.6 Data-Improvement Plan

> The team-photo dataset is a previously evaluated external benchmark. Future development should use new development data, with final performance assessed on a new independent test set after model and threshold choices are fixed.

4. Reproducibility

Item	Technical Specification
Repository	`https://github.com/Yasser-20-alhajlah/smart-pet-vision`
Evidence & Artifacts	**Public evidence:** Checkpoints are available in `Phase1_Checkpoints.zip`. Saved data splits, classification results and checkpoint-verification exports are available in `Phase1_Results_v2.zip` via GitHub Releases (phase1-v1.0). Restore its `data/`, `classification/` and `verification_2026-10-28/` folders under `smart-pet-vision/runs/`. Recomputing metrics from saved predictions, fresh checkpoint inference and full training reproduction are distinct procedures.
Configuration	`configs/config.yaml` (unified configuration for all pipelines)
Environment	`requirements.txt` with pinned package versions

5. Self-Assessment

> **Team self-assessment only. These scores are not awarded grades and do not guarantee full compliance with the official grading rubric.**

Criterion	Max Points	Score	Core Justification
M1: Data Foundations	12	12	Group-stratified split; leak-free duplicates; rigorous audits. Identity leakage analyzed via similarity sensitivity; animal identity labels unavailable.
M2-M3: Classification	20	20	Complete metric reporting; McNemar test; ablation across 3 setups. Temperature calibration bounded by near-separable validation set.
Reproducibility	8	8	Public GitHub release; pinned dependencies; unified config. Large

Appendix A: Dataset Card Summary

(Corresponds to `docs/dataset_card.md` in repository)

Dataset Origin & License

- **Oxford-IIIT Pet:** Parkhi et al. (2012). 37 breeds (12 cat, 25 dog); species and breed labels, pixel trimaps. Train/val/test usage. License: CC BY-SA 4.0.
- **Team-collected:** 401 photos of cats and dogs. Supplied by team members (104 taken by pet owners), annotated by the team, owner consent. External test only. Not redistributed.

Cleaning & Preprocessing

- Unreadable files, images under 32 px, and images without a usable trimap dropped: 14 dropped (empty trimap).
- Exact duplicates (MD5): 0 found. Near-duplicates (pHash Hamming distance ≤ 4): 45 pairs, merged transitively into 41 groups covering 84 images.
- **After cleaning:** 7,335 images (2,366 cats, 4,969 dogs, 37 breeds).

Leakage & Splits

Split 70/15/15 by image count, stratified by breed, whole duplicate groups assigned to one split.

> **Identity-leakage limitation:** Duplicate groups do not cross the train, validation and test splits. However, animal identities are unavailable, so identity-disjointness is not established. Image-similarity filtering is a sensitivity analysis, not a measurement or causal upper bound on leakage.

Known Limitations

- Mostly one large, centred animal per image; web and pet-site photos are cleaner than shelter photos, so small or partly visible animals are under-represented.
- Class imbalance: about two dogs per cat; handled by breed stratification and macro-F1, no class weights.
- Possible overlap between ImageNet pretraining images and Oxford-IIIT Pet (both web-collected) cannot be excluded.

Privacy and Storage (Team Photos)

The processed set (images, masks, annotations, label and audit files) is kept outside the code repository in `SmartPetVision_TeamData`, shared only within the team and, for grading, with the instructor. Unprocessed originals, which may carry camera metadata, are kept privately and are never shared. No team photo file is stored in the repository or redistributed.

Audit Summary

- **Species audit:** 0 errors observed in 200 random images.
- **Bounding Box audit:** 1.3% loose boxes derived from trimaps (4 out of 300).